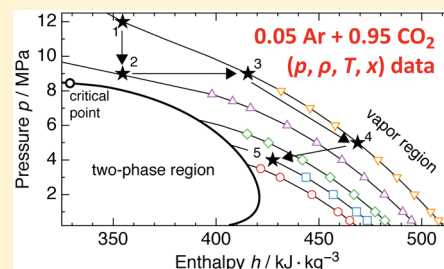


Vapor-Phase (p, ρ, T, x) Behavior and Virial Coefficients for the Binary Mixture (0.05 Argon + 0.95 Carbon Dioxide) over the Temperature Range from (273.15 to 323.15) K with Pressures up to 9 MPa

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ABSTRACT: Accurate density measurements on a binary (argon + carbon dioxide) mixture were carried out at temperatures $T = (273.15, 283.15, 293.15, 308.15, \text{ and } 323.15)$ K with pressures up to the dew-point pressure or 9.0 MPa, whichever was lower. A two-sinker magnetic suspension densimeter was utilized for the measurements. The composition of the gravimetrically prepared mixture was 0.94951 mole fraction carbon dioxide. Taking the measurement uncertainties in temperature, pressure, density, and composition into account, the relative combined expanded uncertainty ($k = 2$) in density was estimated to be less or equal 0.043% of the measured density value. Relative deviations of the new experimental data from the GERG-2008 equation of state (EOS) and from the EOS-CG (a recently developed multiparameter EOS optimized for combustion gases) were within 0.95% and 0.18%, respectively. Third-order virial equations were fitted to the new experimental data. The correlated molar masses at different isotherms agreed with the gravimetrically determined one within 0.02%. Values and uncertainties of the second and third virial coefficients and the second interaction virial coefficient for the binary mixture were determined.



1. INTRODUCTION

Among all the strategies in mitigating global carbon emissions, carbon capture and storage (CCS) was considered to be very efficient and practical. Significant components of the carbon dioxide-rich mixtures involved in the pipeline transportation in CCS are carbon dioxide, argon, nitrogen, oxygen, water, sulfur oxide, and nitric oxide.¹ Volumetric properties of the carbon dioxide-rich mixtures are fundamental parameters in the process design for CCS. Such properties can be calculated conveniently with multiparameter Helmholtz equations of state (EOS) for mixtures; for example, the GERG-2008 equation^{2,3} for natural gas or similar mixtures and the recently developed EOS-CG^{4,5} optimized for combustion gases. These equations are based on high-accuracy equations of state for the mixture components together with departure functions, reducing functions, or combining rules for the binary constituents of a mixture. However, to the best of our knowledge, the situation of accurate vapor-phase (p, ρ, T, x) data for binary carbon dioxide-rich mixtures is rather poor. In our present research project, we investigated the binary system (carbon dioxide + argon); the available experimental (p, ρ, T, x) data for this particular system were summarized by Yang et al.⁶ No data are available with compositions of about 0.95 mole fraction carbon dioxide. To fill this data gap, accurate density measurements at temperatures $T = (273.15, 283.15, 293.15, 308.15, \text{ and } 323.15)$ K with pressures up to dew-point pressure or $p = 9.0$ MPa, whichever is lower, were carried out on a (0.05049 argon + 0.94951 carbon dioxide) mixture utilizing a two-sinker magnetic suspension densimeter.

2. EXPERIMENTAL SECTION

2.1. Apparatus Description. The measurements reported here were carried out with a two-sinker magnetic suspension densimeter operable over the temperature range from (273.15 to 323.15) K at pressures up to 9 MPa; it was described in detail in previous papers^{7,8} and was overhauled for recent studies by our group.^{9,10} The full particulars of this general type of instrument are reported elsewhere.^{11–14} Thus, we just briefly explain the measuring principle here.

Two specially matched sinkers were used for density measurement: a hollow sphere ($V_S \approx 107 \text{ cm}^3$; $m_S \approx 123 \text{ g}$; $\rho_S \approx 1.16 \text{ g} \cdot \text{cm}^{-3}$) and a solid ring ($V_R \approx 15.6 \text{ cm}^3$; $m_R \approx 123 \text{ g}$; $\rho_R \approx 7.90 \text{ g} \cdot \text{cm}^{-3}$). Both sinkers were made of stainless steel and had approximately the same surface area. To obtain a very smooth surface finish, their surfaces were polished electrolytically and afterward gold plated. To measure the density, both sinkers were weighed alternately with an analytical balance (readability: 0.00001 g) while they were immersed in the gas of unknown density. The sinkers had nearly the same mass, the same surface area, and the same surface material but had a substantial difference in volume ($V_S - V_R \approx 91.4 \text{ cm}^3$). Accordingly, the buoyancy effect was very large, and therefore, even the low-density range of gases could be measured accurately. The density of the gas under study was determined by

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$$\rho_{\text{gas}}(T, p) = \frac{(m_S - m_R) - (m_{S,\text{fluid}}^* - m_{R,\text{fluid}}^*)}{V_S(T, p) - V_R(T, p)} \quad (1)$$

where m and V are the mass and the volume of the sinker, m^* is the weighing value of a sinker being surrounded by a gas, and the subscripts refer to the two sinkers. The difference between the masses of the two sinkers is very small, $(m_S - m_R) \approx 0.00080$ g. To improve the accuracy of the value $(m_{S,\text{fluid}}^* - m_{R,\text{fluid}}^*)$, the sinkers were usually exchanged 30 times, and an average value of 30 single values was calculated. (Please note: The balance was tared to $m_{S,\text{fluid}}^* = 0.00000$ g when the spherical sinker was weighed for the first time during a density-measurement run.) A magnetic suspension coupling was used to transmit the gravity and buoyancy forces on the sinkers to the balance, thus isolating the gas sample (inside the pressure tight measuring cell) from the balance that was placed under ambient conditions. The force transmission error (FTE) of the magnetic suspension coupling¹⁵ was very small due to the use of nonmagnetic materials of the coupling housing and because of a nearly identical position of the permanent magnet of the magnetic suspension coupling for both sinker weighings (two-sinker principle). However, a small fluid-specific effect still existed. This remaining error was covered by the uncertainty in density measurement.

2.2. Experimental Material. The (argon + carbon dioxide) gas mixture was prepared gravimetrically in a 20 L aluminum gas cylinder of Scott Specialty Gases with a special treatment of the inside surface, which was intended to improve the long-term stability of the mixture composition. The pure components we used to prepare the mixture are described in Table 1. The materials were used as received, no further gas

Table 1. Sample Information

chemical name	source	purity/mol fraction	purification method
argon	Air Liquide	0.999990 ^a	none
carbon dioxide	Air Liquide	0.999995 ^b	none

^aImpurities (stated by supplier): $x(\text{H}_2\text{O}) \leq 2.0 \cdot 10^{-6}$, $x(\text{O}_2) \leq 2.0 \cdot 10^{-6}$, $x(\text{C}_m\text{H}_n) \leq 0.5 \cdot 10^{-6}$, $x(\text{CO}_2) \leq 0.2 \cdot 10^{-6}$, $x(\text{N}_2) \leq 5.0 \cdot 10^{-6}$.

^bImpurities (stated by supplier): $x(\text{H}_2\text{O}) \leq 2.0 \cdot 10^{-6}$, $x(\text{O}_2) \leq 1.0 \cdot 10^{-6}$, $x(\text{C}_m\text{H}_n) \leq 0.1 \cdot 10^{-6}$, $x(\text{N}_2) \leq 2.0 \cdot 10^{-6}$, $x(\text{CO}) \leq 0.5 \cdot 10^{-6}$, $x(\text{NO}_x) \leq 0.1 \cdot 10^{-6}$.

analysis was conducted to confirm their purity. We prepared the gas mixture with a comparatively high filling pressure of the sample cylinder of approximately 14 MPa. The concentrations were determined to be 0.05049 mole fraction argon and 0.94951 mole fraction carbon dioxide, respectively. The molar mass of the gas mixture was $M_{\text{grav}} = 43.8047$ g·mol⁻¹. The expanded uncertainty ($k = 2$) in composition was estimated to be 0.0004 mole fraction, which corresponds to an expanded uncertainty ($k = 2$) in molar mass of 0.0016 g·mol⁻¹.

A detailed description of our sample preparation procedure and the mixture-preparation system that we used can be found in section 6 of the dissertation of Schäfer¹⁶ who recently finished his doctorate in our institute.

2.3. Experimental Procedures. Density measurements were carried out along isotherms at $T = (273.15, 283.15, 293.15, 308.15, \text{ and } 323.15)$ K. For temperatures below the maxcondentherm ($T = 303.08$ K), the pressures extended to the dew-point pressures, while for those above the maxcondentherm, the pressures extended up to 9.0 MPa. The measured points are depicted in Figure 1 in a pressure versus

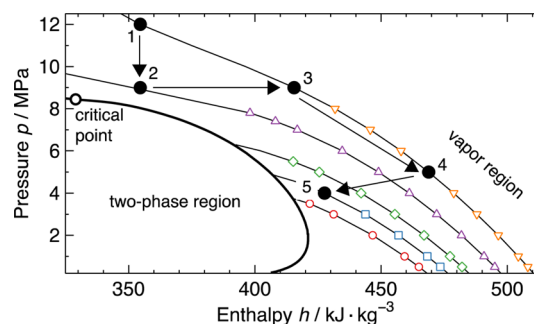


Figure 1. Pressure–enthalpy phase diagram of the (0.05049 argon + 0.94951 carbon dioxide) mixture. Density values measured in the present work: ○, $T = 273.15$ K; □, $T = 283.15$ K; ◇, $T = 293.15$ K; △, $T = 308.15$ K, and ▽, $T = 323.15$ K. The solid curve represents the liquid–vapor phase boundary. The solid points (●) illustrate an example for the experimental procedure (with the help of arrows) to attain the first measurement point (5) at $T = 283.15$ K, where state point (1) indicates the condition of the gas mixture inside the sample cylinder; see explanations in section 2.3.

enthalpy phase diagram, which was calculated with the GERG-2008 equation of state for natural gas mixtures of Kunz and Wagner^{2,3} as implemented in the NIST REFPROP database of Lemmon et al.¹⁷

Several measurements were conducted close to the dew line. To avoid a possible phase transition due to isenthalpic expansion of the gas from the sample cylinder into the two-phase region, an elaborate experimental procedure was carried out as indicated by the arrows in Figure 1 for measurements along the $T = 283.15$ K isotherm as an example. At first, the gas mixture inside the sample cylinder was in a supercritical state (Point 1, $T \approx 323.15$ K and $p \approx 12$ MPa). The cylinder was heated at the bottom to create convection inside the cylinder for gas mixing. The measuring cell and its connection tubes to the gas-dosing system were heated and controlled at $T = 323.15$ K, and the filling line from the sample cylinder to the gas-dosing system was at ambient temperature. Then, the pressure regulator at the sample cylinder was adjusted to $p \approx 9.0$ MPa, which was the maximum pressure for the present measurements. Thus, the pressure was reduced in an isenthalpic way as shown from point (1) to point (2) in Figure 1. Subsequently, the evacuated measuring cell was filled; the change of state from point (2) to (3) in Figure 1. (Please note: When filling the evacuated measuring cell, the gas unavoidably attained conditions in the two-phase region. However, once the gas was in point (3), the cell was flushed supercritically with fresh gas, thus ensuring that the composition of the first filling inside the cell was correct.) About 30 min after flushing the measuring cell, which was at $T = 323.15$ K, the temperature of the cell was re-equilibrated, and the pressure was slightly higher than 9.0 MPa. Before attaining state point (4), the density of the first intended measuring point (5) at $T = 283.15$ K and $p \approx 4.0$ MPa was calculated to enable an isochoric change of state from point (4) to (5). With the density as known input parameter, a starting pressure of 5.0 MPa was calculated for a temperature of 323.15 K. Thus, the isothermal change of state from point (3) to (4) could be realized by venting gas and slowly decreasing the pressure from about 9 MPa to 5 MPa, (Figure 1). For the isochoric change of state from point (4) to (5), the temperature of the measuring cell was slowly reduced to $T = 283.15$ K. After allowing enough time for equilibration, density measurements were started at $p = 4.0$ MPa followed by measurements at lower

pressures, i.e., $p = (3.0, 2.0, 1.0, 0.5)$ MPa. Isothermal measurements were conducted with a single filling over several days because multiple replicates at the same (T, p) state point were taken to investigate the influence of sorption effects, which especially occur when measuring near the dew line. At each (T, p) state point, at least two replicate measurements were conducted.

Because of isothermal measurements with only a single filling of the measuring cell and measurements in the vicinity to the dew line, it was known that the mixture composition could vary due to sorption effects.⁹ Carbon dioxide is the component that would preferentially adsorb on the solid surfaces inside the measuring cell, and it would subsequently desorb from them at low pressures. Moreover, carbon dioxide is known for its affinity to diffuse into elastomer seals. The consequence of both is a change in composition of the measured gas and, therefore, a change of the gas density inside the measuring cell. Richter and Kleinrahm⁹ suggested to investigate sorption effects for thermophysical property measurements on mixtures and demonstrated that flushing the measuring cell before measurements are conducted was an effective way to obtain accurate results. However, in this work, it was only possible to flush the measuring cell at $T = 323.15$ K and $p \approx 9.0$ MPa (point (3) in Figure 1) due to the course of the phase boundary. Therefore, flushing the cell was carried out before moving to the first measuring point on each isotherm.

3. RESULTS AND DISCUSSION

3.1. Uncertainty in Measurement. We used the “Guide to the Expression of Uncertainty in Measurement” (commonly known as GUM) according to ISO/IEC (2008)¹⁸ for the determination of the measurement uncertainty and assumed for the present analysis that the input quantities are not correlated.

To determine the combined uncertainty in density, we considered the following input quantities: (1) The standard uncertainty in density measurement was estimated to be 0.0075% of the measured value (incl. the fluid-specific effect of the FTE), which was valid for the entire temperature range and pressure range of the densimeter. (2) The standard uncertainty in temperature measurement was estimated to be 2.5 mK. (3) The standard uncertainty in pressure measurement was estimated to be 0.0035% of the measured pressure. (4) The standard uncertainty in density arising from the uncertainty in composition (both mixture preparation and sorption effects) was determined according to a recent publication of Richter and McLinden.¹⁹ The standard uncertainty in mixture composition (from gravimetry) was ascertained to be 0.0002 mole fraction (see section 2.2.), which corresponds to a relative standard uncertainty in molar mass of 0.002%. Moreover, a virial equation was fitted to the experimental data along each isotherm, and a molar mass M_{VE} was determined (see section 3.3). The standard deviation of the differences between M_{VE} and the gravimetrically determined molar mass M_{grav} was 0.01%. (5) To check the reproducibility of the measurements, one of the isotherms was measured three times. Moreover, from replicate measurements over time along each isotherm we could quantify density shifts due to adsorption and desorption effects. From these results, we estimated an additional standard uncertainty of 0.012% of the measured density.

Taking the mentioned uncertainties into account, the relative combined expanded uncertainty ($k = 2$) in density was estimated to be less than or equal to 0.043% for the values measured within the present work. The relative combined

expanded uncertainties ($k = 2$) are listed for each state point in Table 2.

3.2. Comparison to Equations of State. The experimental (p, ρ, T) data are listed in Table 2. Relative deviations of the isothermal experimental data from values calculated with

Table 2. Experimental (p, ρ, T) Data for the (0.05049 Argon + 0.94951 Carbon Dioxide) Mixture and Relative Deviations of Experimental Densities ρ from Densities Calculated with the GERG-2008 Equation of State^{2,3} and the EOS-CG,^{4,5} Where p is the Pressure and T is the Temperature (ITS-90) and $U_c(\rho)/\rho$ is the Relative Combined Expanded Uncertainty ($k = 2$) in Density^a

p/MPa	$\rho/(\text{kg}/\text{m}^3)$	$100U_c(\rho)/\rho$	$100(\rho - \rho_{\text{GERG}})/\rho_{\text{GERG}}$	$100(\rho - \rho_{\text{EOS-CG}})/\rho_{\text{EOS-CG}}$
$T = 273.150$ K				
2.94815	72.9950	0.039	0.3980	0.0420
2.52158	59.4639	0.039	0.2861	0.0191
2.05459	46.2540	0.038	0.2027	0.0109
1.04568	21.6518	0.038	0.0871	0.0097
0.52654	10.5097	0.038	0.0524	0.0174
$T = 283.150$ K				
4.05842	105.393	0.039	0.4560	−0.0476
3.05411	70.7119	0.039	0.2139	−0.0645
2.04559	43.4280	0.038	0.1074	−0.0398
1.03404	20.4562	0.038	0.0520	−0.0089
0.48717	9.32016	0.038	0.0308	0.0048
$T = 293.150$ K				
5.53047	157.768	0.040	0.8358	0.0563
5.04518	133.561	0.040	0.5389	−0.0451
4.03274	95.0463	0.039	0.2699	−0.0749
3.03107	65.5032	0.038	0.1532	−0.0531
2.03346	40.9568	0.038	0.0837	−0.0305
1.00989	19.1313	0.038	0.0346	−0.0129
0.50122	9.23865	0.038	0.0138	−0.0079
$T = 308.150$ K				
7.79344	248.529	0.043	0.9502	−0.0573
7.41835	221.117	0.041	0.7574	−0.0960
6.99178	195.441	0.041	0.6114	−0.0977
6.00531	149.364	0.039	0.3834	−0.0904
5.01295	113.924	0.039	0.2505	−0.0708
4.00582	84.4871	0.039	0.1666	−0.0478
3.00453	59.5083	0.038	0.1104	−0.0266
2.00249	37.5504	0.038	0.0732	−0.0054
0.99602	17.7876	0.038	0.0384	0.0046
0.49391	8.62503	0.038	0.0273	0.0116
$T = 323.150$ K				
8.98797	249.635	0.041	0.3766	−0.1775
8.53345	226.233	0.040	0.3415	−0.1746
8.02380	202.754	0.040	0.2885	−0.1754
7.03266	163.762	0.039	0.2289	−0.1334
6.03243	130.908	0.039	0.1763	−0.0977
5.04562	103.062	0.039	0.1328	−0.0700
4.03033	77.9178	0.038	0.0962	−0.0470
3.01239	55.4385	0.038	0.0672	−0.0276
2.00983	35.3997	0.038	0.0428	−0.0134
1.03015	17.4463	0.038	0.0227	−0.0028
0.52121	8.65936	0.038	0.0114	−0.0007

^aThe expanded uncertainties ($k = 2$) of measurements are 5 mK for temperature, 0.007% for pressure, and 0.015% for density. The mixture sample was prepared gravimetrically with the expanded uncertainty ($k = 2$) in composition 0.0004 mole fraction.

the GERG-2008 equation of state^{2,3} (top diagram) and the EOS-CG (bottom diagram) are shown in Figure 2. The EOS-

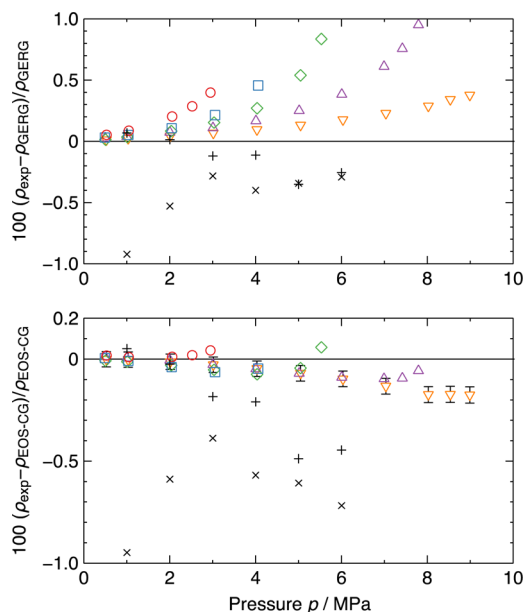


Figure 2. Relative deviations of experimental densities ρ_{exp} for the (0.05049 argon + 0.94951 carbon dioxide) mixture from densities ρ_{GERG} calculated with the GERG-2008 equation of state^{2,3} (zero line in top diagram) and densities $\rho_{\text{EOS-CG}}$ calculated with the EOS-CG^{4,5} (zero line in bottom diagram). Density values measured in the present work: $\circ$, $T = 273.15$ K; $\square$, $T = 283.15$ K; $\diamond$, $T = 293.15$ K; $\triangle$, $T = 308.15$ K and ∇ , $T = 323.15$ K. Relative deviations of experimental densities ρ_{exp} for the (0.0308 argon + 0.9692 carbon dioxide) mixture measured by Mantovani et al.¹⁵ are plotted for comparison: $\times$, $T = 303.22$ K; $+$, $T = 323.18$ K. For clarity, error bars for the relative combined expanded uncertainty ($k = 2$) in the density determination are illustrated only for the measurements at $T = 323.15$ K (bottom diagram); in the top diagram the error bars essentially correspond to the vertical size of the plot symbol.

CG^{4,5} is a relatively new Helmholtz energy model for CCS mixtures and was used as implemented in the TREND software package²⁰ of Ruhr-University Bochum. For the vapor phase of the binary gas mixture investigated in the present work, the authors of the GERG equation report a relative uncertainty of (0.5 to 1.0) %, and the authors of the EOS-CG report for the

(argon + carbon dioxide) system a relative uncertainty of less than 1.0%.

Generally, the relative deviations decrease with decreasing pressure along each isotherm and, as expected from theory, converge to zero when approaching the ideal gas (at low pressures). Comparing our measured densities with the GERG-2008 equation, obvious positive deviations can be observed. The relative deviations increase with decreasing temperature and increasing pressure, reaching a maximum of 0.95% at $T = 308.15$ K and $p = 7.8$ MPa. In contrast, relative deviations of our experimental data from the EOS-CG are less than 0.18%, thus, being significantly smaller compared to the GERG equation. The reason is that the EOS-CG employs a binary specific departure function for the (argon + carbon dioxide) system, whereas the GERG-2008 does not.

Experimental densities of Mantovani et al.²¹ for the binary mixture (0.0308 argon + 0.9692 carbon dioxide) at $T = (303.22$ and $323.18)$ K with pressures up to 6 MPa are also plotted in Figure 2. The agreement with both equations of state is rather poor. No systematic trend of the relative deviations can be observed, and the data do not converge to zero when approaching the ideal gas. These data and the four (p , ρ , T , x) data of Sarashina et al.²² for the binary mixture (0.071 argon + 0.929 carbon dioxide) at $T = 288.15$ K (ITS-68) with pressures between (2.43 and 4.93) MPa are the only data for the present binary system in the studied ranges of composition, temperature, and pressure. However, the relative deviations of the data of Sarashina et al.²² from the equations of state range between (0.73 and 1.79) %, thus, we decided to not plot the data in Figure 2.

3.3. Virial Coefficients. The third-order virial equation

$$p = \frac{\rho R_m T}{M} \left(1 + B(T) \cdot \frac{\rho}{M} + C(T) \cdot \left(\frac{\rho}{M} \right)^2 \right) \quad (2)$$

was fitted to our isothermal data, where p is the pressure, T is the temperature, ρ is the density, M is the molar mass of the gas mixture, $R_m = 8.314472 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ is the molar gas constant, and $B(T)$ and $C(T)$ are the second and the third virial coefficients, respectively. All data were weighted equally, and the results are given in Table 3. Attempts to include the fourth virial coefficient did not substantially improve the goodness of the fit for data with pressures lower than p_{lim} . Therefore, data with pressures higher than p_{lim} , for example, $p_{\text{lim}} = 7.1$ MPa at $T = 323.15$ K as listed in Table 3, were excluded from the fitting

Table 3. Summary of the Virial Analysis for the (0.05049 Argon + 0.94951 Carbon Dioxide) Mixture^a

T K	p_{lim} MPa	M is an unknown parameter				$M_{\text{grav}} = 43.8047 \text{ g/mol}$						
		RMS_{VE} %	M_{VE} (g/mole)	ΔM (g/mole)	$x(\text{CO}_2)$ (mol frac)	RMS_{grav} %	B (cm ³ /mol)	$U(B)$ (cm ³ /mol)	C (cm ³ /mol) ²	$U(C)$ (cm ³ /mol) ²	B_{12} (cm ³ /mol)	$U(B_{12})$ (cm ³ /mol)
273.150	3.0	0.0010	43.8140	0.0093	0.95179	0.0064	-141.03	0.40	5055	207	-57.4	5.1
283.150	4.1	0.0010	43.8105	0.0058	0.95092	0.0053	-129.46	0.27	4835	100	-48.3	3.9
293.150	5.6	0.0005	43.8024	-0.0023	0.94894	0.0022	-119.59	0.17	4680	43	-44.8	3.0
308.150	7.5	0.0025	43.8085	0.0038	0.95043	0.0046	-106.67	0.11	4399	21	-40.7	2.5
323.150	7.1	0.0005	43.8061	0.0014	0.94984	0.0014	-95.26	0.17	4026	41	-34.1	2.6

^a M is the molar mass; p_{lim} is the pressure limit above which experimental data were not used for fitting the virial equation; RMS is the root-mean-square of the relative deviations in pressure of the experimental data from values calculated with the fitted virial eq (eq 2); M_{VE} is the molar mass determined by the virial analysis (eq 2); ΔM is the absolute difference of the fitted molar mass M_{VE} from the gravimetrically determined one ($M_{\text{grav}} = 43.8047 \text{ g/mol}$); $x(\text{CO}_2)$ is the mole fraction of carbon dioxide determined with the fitted molar mass M_{VE} ; B and $U(B)$ are the second virial coefficient and its expanded uncertainty ($k = 2$), respectively; C and $U(C)$ are the third virial coefficient and its expanded uncertainty ($k = 2$), respectively; B_{12} and $U(B_{12})$ are the second interaction virial coefficient calculated by eq 3 and its expanded uncertainty ($k = 2$), respectively.

procedure because it was not reasonable to fit the third-order virial equation to higher densities.

At first, eq 2 was fitted to the isothermal data with the molar mass M of the mixture as an unknown parameter, and excellent correlations were obtained. Initially, this was done to check on the gravimetric composition but mainly it was an important indicator of sorption effects (see below). The root-mean-squares (RMS_{VE}) of the relative deviations in pressure of the experimental data from values calculated with the fitted virial equation were smaller than 0.0025% for all isotherms (see Table 3). Relative deviations of the five correlated molar masses M_{VE} from the gravimetrically determined one ($M_{\text{grav}} = 43.8047 \text{ g/mol}$) of up to about 0.02% were observed, which was larger than the combined uncertainty ($k = 2$) of 0.004% of M_{grav} . The reasons were that (1) the difference in molar masses of carbon dioxide and argon ($M_{\text{CO}_2} = 44.0098 \text{ g}\cdot\text{mol}^{-1}$ and $M_{\text{Ar}} = 39.9480 \text{ g}\cdot\text{mol}^{-1}$) was small, (2) not enough data could be fitted, especially at $T = (273.15 \text{ and } 283.15) \text{ K}$, and (3) adsorption and desorption effects probably distorted the composition. In a second step, $B(T)$ and $C(T)$ in eq 2 were fitted with the molar mass M_{grav} being a constant. Thereby, the root-mean-squares (RMS_{grav}) of the relative deviations in pressure became slightly larger but were still smaller than 0.0064% for all isotherms (see Table 3). The fitted values for $B(T)$ and $C(T)$ are listed in Table 3 together with the second interaction virial coefficient $B_{12}(T)$ as defined by

$$B_{12}(T) = \frac{B(T) - x_1^2 B_{11}(T) - x_2^2 B_{22}(T)}{2x_1 x_2} \quad (3)$$

where x_i is the mole fraction of component i , $B_{ii}(T)$ is the second virial coefficient of the pure gas i , and the subscripts 1 and 2 represent carbon dioxide and argon, respectively. $B_{ii}(T)$ was determined from the respective reference equations of state for argon²³ and carbon dioxide.²⁴ The uncertainties of the virial coefficients were also calculated (see Table 3). The main contributions to the uncertainty of $B(T)$ and $C(T)$ were the statistical uncertainties of fitting the virial equation, the uncertainty of the measurements, and the uncertainty of M_{grav} . The uncertainty of $B_{12}(T)$ was determined by applying the error propagation principle to eq 3. As can be seen in Table 3, both $B(T)$ and $B_{12}(T)$ increase with increasing temperature while $C(T)$ decreases with increasing temperature, which is a plausible result.

In Figure 3, we plot the relative deviations of the experimental densities from densities calculated with the virial model using the fitted virial coefficients $B(T)$ and $C(T)$ as listed in Table 3. The virial equation as given in eq 2 represents the experimental densities up to pressures $p = p_{\text{lim}}$ (see Table 3) within 0.013%, which is considerably less than the standard uncertainty of about 0.02% of the measured densities.

4. CONCLUSIONS

We reported accurate results of (p, ρ, T) measurements on a gravimetrically prepared (0.05049 argon + 0.94951 carbon dioxide) mixture over the temperature range from $T = (273.15 \text{ to } 323.15) \text{ K}$ with pressures up to 9.0 MPa or the dew-point pressure, whichever was lower. A two-sinker magnetic suspension densimeter was utilized for the measurements. The relative combined expanded uncertainty ($k = 2$) in density was estimated to be less than or equal to 0.043%. The GERG-2008 equation of state and the EOS-CG represent the experimental data within about 1.0% and 0.2%, respectively,

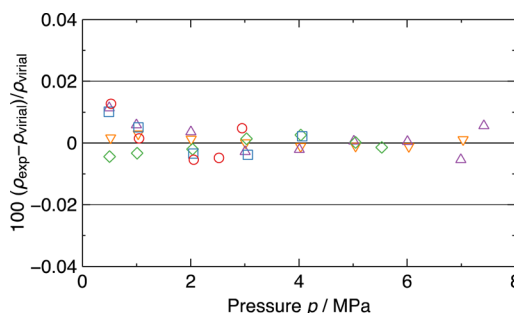


Figure 3. Relative deviations of experimental densities ρ_{exp} for the (0.05049 argon + 0.94951 carbon dioxide) mixture from densities ρ_{virial} calculated with the virial model as given in eq 2 (zero line). Density values measured in the present work: $\circ$, $T = 273.15 \text{ K}$; $\square$, $T = 283.15 \text{ K}$; $\diamond$, $T = 293.15 \text{ K}$; $\triangle$, $T = 308.15 \text{ K}$ and ∇ , $T = 323.15 \text{ K}$. Values are only plotted for pressures below p_{lim} (the experimental data above p_{lim} were not used for fitting the virial equation, see Table 3); for these values the relative combined expanded uncertainty ($k = 2$) in density was estimated to be less than or equal to 0.041% (see Table 2).

which is within the stated uncertainties of the equations. The isothermal experimental data were also used to fit third-order virial equations in order to determine the second and third virial coefficients. In summary, the new experimental data are very accurate and could be used together with the virial coefficients that we determined to further improve the performance of the multiparameter equations of state for mixtures.

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Notes

The authors declare no competing financial interest.

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